

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – II
PAPER –2
TEST DATE: 30-06-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

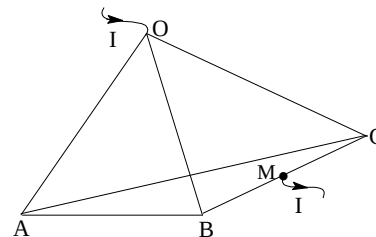
Physics

PART – I

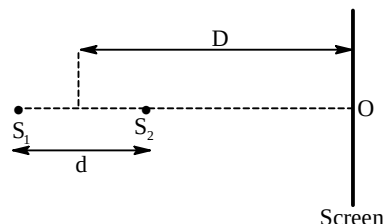
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

- Suppose we make a frame of a regular tetrahedron using six resistors with the same resistance, r and apply a voltage V between point O and point M as shown in the figure. A current I flows into point O and flows out of point M . M is the midpoint of resistor BC .
 - The current between O and B is $3/8I$
 - The current between A and B is $I/8$
 - The potential at point B is $2V/5$
 - The equivalent resistance between point O and M is $5r/8$



- Two point monochromatic and coherent sources of light of wavelength λ are placed as shown in figure. The initial phase difference between the sources is 0. $D \gg d$. Mark the **CORRECT** statement(s).



- If $d = \frac{7\lambda}{2}$, O will be minima
 - If $d = \lambda$, only one maxima can be observed on screen
 - If $d = 4.8\lambda$, then total 10 minima would be there on screen.
 - If $d = \frac{5\lambda}{2}$, then intensity at O would be minimum.
- The two switches in the circuit shown in figure have been closed for a long time. At $t = 0$, switch 1 is opened. Then 35 ms later, switch 2 is opened:
 - The current through the inductor for $t < 35\text{ms}$ is $6e^{-40t}\text{A}$
 - The current through the inductor for $t \geq 35\text{ms}$ is $1.48e^{-60(t-0.035)}\text{A}$
 - Heat lost in the 18Ω resistor is 845.27 mJ .
 - Heat lost in the 3Ω resistance is 618.24 mJ .
 - At the center of a hollow hemisphere uniformly charged with a surface charge density σ , there is a small dipole of dipole moment ρ . The moment of inertia about the rotation axis is I .
 - The magnitude of $\vec{E}$ due to hemisphere at the center of hemisphere is $\sigma/2\epsilon_0$
 - The minimum potential energy of dipole is $-\rho \frac{\sigma}{4\epsilon_0}$

- (C) If dipole is released from rest from a position perpendicular to the symmetry axis, the maximum KE of dipole is $\frac{\sigma\rho}{4\epsilon_0}$
- (D) The dipole performs SHM for small angular displacements with angular frequency is $\omega = \sqrt{\frac{\rho E}{I}}$
5. A spherical shell of radius a is just filled with liquid of density ρ and the system rotates with uniform angular velocity ω about vertical diameter.
- (A) The level from center at which the pressure on the shell is maximum is g/ω^2 .
- (B) The level from center at which the pressure on the shell is maximum is $2g/\omega^2$
- (C) The thrust force due to liquid on the lower half of the shell is $\frac{5}{3}\rho g\pi a^3 + \frac{1}{4}\rho\pi\omega^2 a^4$
- (D) The thrust force on upper half of the shell is $\frac{1}{3}\rho g\pi a^3 + \frac{1}{4}\pi\rho\omega^2 a^4$
6. Let the energy, magnitude of linear momentum and angular frequency of an electron in hydrogen atom be E , p and ω respectively. If n be the corresponding quantum number then:
- (A) $\left(\frac{E}{\omega}\right)$ varies as n .
- (B) $\left(\frac{Ep}{\omega}\right)$ is independent of n .
- (C) $(p\omega)$ varies as $n^{1/2}$
- (D) $(Ep\omega)$ is independent of n .

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

In a homogeneous, non-magnetic, highly insulating and viscous medium, a moving particle experiences a viscous drag given by the law $\vec{f} = -b\vec{v}$. Here b is a positive constant. A particle having charge q projected with an unknown velocity from a point in the medium almost stops (its speed becomes practically negligible) after travelling a distance of 10 m in a straight line. Now a uniform magnetic field is established in the region and the same particle is again launched with the same projection velocity directed perpendicular to the magnetic field.

7. In the presence of above magnetic field, if the particle almost stops at a point 6m away from the point projection the magnetic field is equal to:
- (A) $\frac{4b}{3q}$
- (B) $\frac{3b}{4q}$

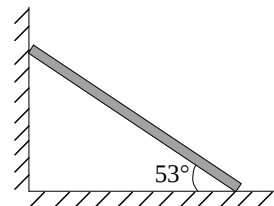
- (C) $\frac{4q}{3b}$
 (D) $\frac{3q}{4b}$

8. The strength of the magnetic field is now doubled in magnitude. How far (in meters) from the point of projection will the particle come to an almost rest?

- (A) $\frac{\sqrt{73}}{2}$
 (B) $\frac{20}{\sqrt{73}}$
 (C) $\frac{40}{\sqrt{73}}$
 (D) $\frac{30}{\sqrt{73}}$

Paragraph for Question Nos. 09 and 10

A rod of length 10 meter has one end on smooth floor and the other end on smooth wall is released from the rest from the position shown in figure. When the rod makes an angle of 37° with the horizontal, answer the following:



9. The angular velocity of rod is:

- (A) $\sqrt{3}$ rad/sec
 (B) $\sqrt{5}$ rad/sec
 (C) $\sqrt{\frac{5}{3}}$ rad/sec
 (D) $\sqrt{\frac{3}{5}}$ rad/sec

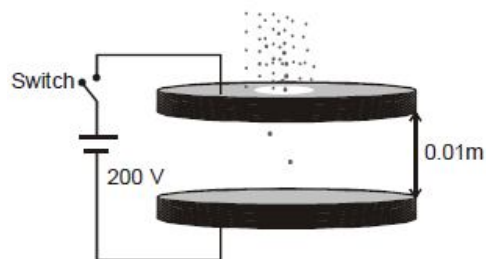
10. The velocity of centre of mass of rod is:

- (A) $\sqrt{5}$ m/s
 (B) $\sqrt{10}$ m/s
 (C) $\sqrt{15}$ m/s
 (D) $\sqrt{20}$ m/s

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

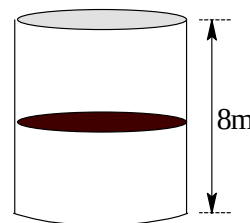
11. Two large circular discs separated by a distance of 0.01 m are connected to a battery via a switch as shown in the figure. Charged oil drops of density 900 kg m^{-3} are released through a tiny hole at the center of the top disc. Once some oil drops achieve terminal velocity, the switch is closed to apply a voltage of 200 V across the discs. As a result, an oil drop of radius $8 \times 10^{-7} \text{ m}$ stops moving vertically and floats between the discs. The number of electrons present in this oil drop is _____.



(Neglect the buoyancy force, take acceleration due to gravity $= 10 \text{ ms}^{-2}$ and charge on an electron $(e) = 1.6 \times 10^{-19} \text{ C}$)

12. A hot air balloon is carrying some passengers, and a few sandbags of mass 1 kg each so that its total mass is 480 kg. Its effective volume giving the balloon its buoyancy is V . The balloon is floating at an equilibrium height of 100 m. When N number of sandbags are thrown out, the balloon rises to a new equilibrium height close to 150 m with its volume V remaining unchanged. If the variation of the density of air with height h from the ground is $\rho(h) = \rho_0 e^{-h/h_0}$, where $\rho_0 = 1.25 \text{ kg m}^{-3}$ and $h_0 = 6000 \text{ m}$, the value of N is _____.

13. A thermally isolated cylindrical closed vessel of height 8 m is kept vertically. It is divided into two equal parts by a diathermic (perfect thermal conductor) frictionless partition of mass 8.3 kg. Thus the partition is held initially at a distance of 4 m from the top, as shown in the schematic figure below. Each of the two parts of the vessel contains 0.1 mole of an ideal gas at temperature 300 K. The partition is now released and moves without any gas leaking from one part of the vessel to the other. When equilibrium is reached, the distance of the partition from the top (in m) will be.



(Take the acceleration due to gravity $= 10 \text{ ms}^{-2}$ and the universal gas constant $= 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$).

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15**Question Stem**

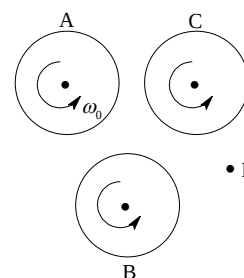
Two small balls have charges q_1 and q_2 and have same velocity. Both of them enter into a uniform electric field simultaneously. After some time the direction of motion of the first ball changes by 60° and its speed get reduced half. The direction of motion of the second ball changes by 90° in the same interval.

14. The speed of second ball reduced by a factor of _____
15. Find the ratio of specific charges of the two balls. Ignore any interaction between the balls.

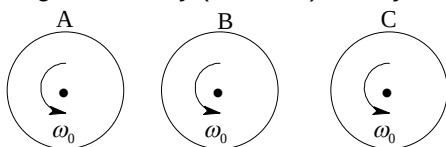
Question Stem for Question Nos. 16 and 17

Question Stem

Three identical cylinders of rough surfaces are mounted on frictionless horizontal axles passing through their centres. They are all spinning about their axles in anti-clockwise direction with same angular speed ω_0 . They are brought closer into contact and their axels are held stationary till slipping between them ceases. [The cylinders are held symmetrically relative to one another as shown, so that all of them touch one another] ($\omega_0 = 12 \text{ rad/sec}$)



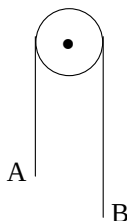
16. Final angular speed (rad/sec) of cylinder B will be _____
17. The arrangement of the three cylinders is changed as shown in figure. Now, what is their final angular velocity (rad/sec) of they are brought into contact?



Question Stem for Question Nos. 18 and 19

Question Stem

A uniform rope of length $L = 2.5 \text{ meter}$ is placed over a smooth pulley and held in a position, where 40% of its length lies on one side and the remaining 60% hangs on the other side. The rope is released from this position. [Neglect dimensions of the pulley] ($g = 10 \text{ m/s}^2$)



18. Find the vertical acceleration (in m/s^2) of the COM of the rope, immediately after it is released.
19. Find the vertical velocity (in m/s^2) of the COM, at the instant end A is $L/5$ above its initial position.

Chemistry

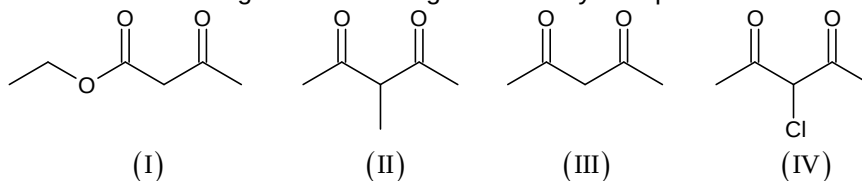
PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

20. Correct statement among the following is:
- Ethoxymethyl chloride reacts with Nucleophiles, faster than 1-Chlorobutane
 - Ethoxymethyl chloride may follow S_N1 mechanism when reacts with Nucleophiles
 - Phenylthio-2-ethyl chloride reacts with water faster than Ethoxymethyl chloride
 - Higher rate of 2-Phenylthioethyl chloride is because the neighbouring group participation by phenyl ring.

21. Correct acidic strength order of the given carbonyl compound is:

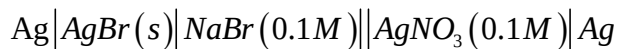


- $IV > I > II > III$
- $IV > II > III > I$
- $IV > III > II > I$
- $IV > I > III > II$

22. Correctly matched among the following is:

Column I (reaction)	Column II (Major product)
(A) $2\text{H}_3\text{C}-\overset{\text{O}}{\underset{\text{ }}{\text{C}}}-\text{CH}_3 \xrightleftharpoons{\text{Ba(OH)}_2}$ (yield is increased in soxhlet apparatus)	$(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{COCH}_3$ Diacetone alcohol
(B) $2\text{CH}_3\text{COCH}_3 \xrightleftharpoons{\text{HCl}}$	$(\text{CH}_3)_2\text{C}=\text{CHCOCH}_3$ Mesityl oxide
(C) $2\text{CH}_3\text{COCH}_3 \xrightleftharpoons[(ii) \text{CH}_3\text{COCH}_3]{(i) \text{HCl}}$	$(\text{CH}_3)_2\text{C}=\text{CHCOCH}=\text{C}(\text{CH}_3)_2$ Phorone
(D) $\text{CH}_3\text{CHO} + \text{CH}_3\text{COCH}_3 \xrightleftharpoons{\text{NaOH}}$	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COCH}_3$ 4-Hydroxypentan-2-one

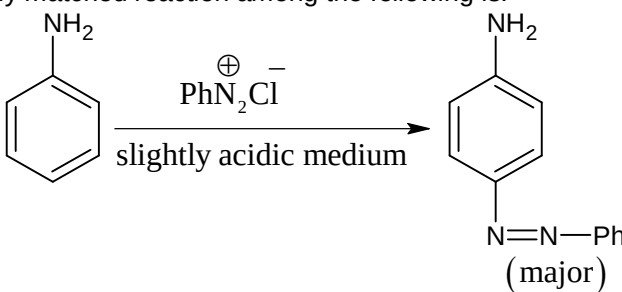
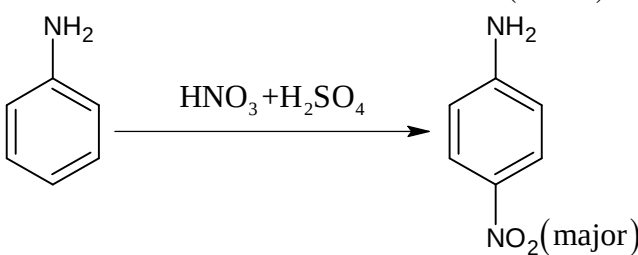
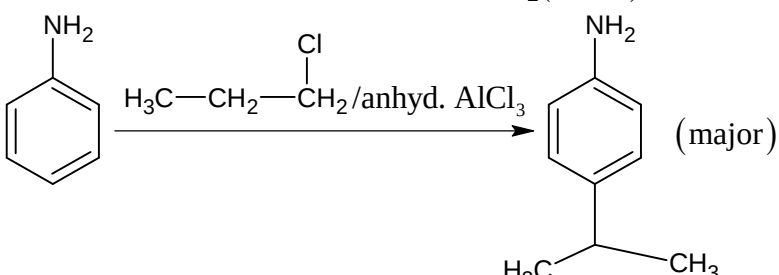
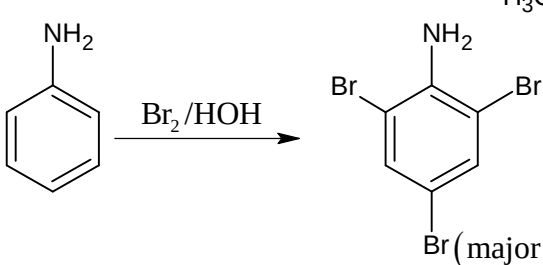
23. The cell shown below generates a potential of 0.6 V at 300 K. Find $(\log K_{sp})$ for AgBr :



$$\left[\text{Take } \frac{2.303RT}{F} = 0.06 \right]$$

- 30.2
- 12
- 12
- 8.9

24. Correctly matched reaction among the following is:

- (A)
- 
- (B)
- 
- (C)
- 
- (D)
- 

25. Correct order of Rate of Reduction among the following is:

- (A) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH} > \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$ -----(Reduction with Borane)
- (B) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H} > \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ -----(Reduction with LiAlH_4)
- (C) $\text{R}-\text{C}\equiv\text{C}-\text{R} > \text{R}-\text{CH}=\text{CH}-\text{R}$ -----(Reduction with H_2/Ni)
- (D) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} > \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ -----(Reduction with DiBAL-H)

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

A metal complex having composition $\text{Cr}(\text{NH}_3)_4\text{Cl}_2\text{Br}$ has been isolated in two forms (A) and (B). The form (A) reacts with AgNO_3 to give a white precipitate readily soluble in dilute aqueous ammonia, whereas (B) gives a pale yellow precipitate soluble in concentrated ammonia. Now given the answers to following questions:

26. The compound (A) and (B) are:
- (A) $[\text{Cr}(\text{NH}_3)_4\text{BrCl}]\text{Cl}$ and $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$
- (B) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$ and $[\text{Cr}(\text{NH}_3)_4\text{BrCl}]\text{Cl}$
- (C) $[\text{Cr}(\text{NH}_3)_3.\text{Cl}_2\text{Br}].\text{NH}_3$ and $[\text{Cr}(\text{NH}_3)_4\text{BrCl}]\text{Cl}$
- (D) $[\text{Cr}(\text{NH}_3)_4.\text{Cl}_2]\text{Br}$ and $[\text{Cr}(\text{NH}_3)_3.\text{Cl}_2\text{Br}].\text{NH}_3$
27. The magnetic moment of A and B are, respectively:
- (A) 3.87 BM, 1.36 BM
- (B) 1.36 BM, 3.87 BM
- (C) 3.87 BM, 3.87 BM
- (D) 1.36 BM, 1.36 BM

Paragraph for Question Nos. 28 and 29

Novocaine, a local anesthetic, is a compound with molecular formula $\text{C}_{13}\text{H}_{20}\text{O}_2\text{N}_2$. It is insoluble in water and dilute NaOH, but soluble in dil. HCl. Upon treatment with NaNO_2 and HCl and then with β -naphthol, a highly coloured solid is formed.

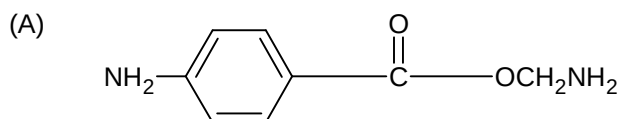
When Novocaine is boiled with aqueous NaOH, it slowly dissolves. The alkaline solution is shaken with ether and layers are separated.

Acidification of the aqueous layer causes the precipitation of white solid 'A'; continued addition of acid causes 'A' to redissolve. Upon isolation 'A' is found to have melting point of $185-186^\circ\text{C}$ and the formula $\text{C}_7\text{H}_7\text{O}_2\text{N}$.

Evaporation of the ether layer leaves a liquid 'B' of formula $\text{C}_6\text{H}_{15}\text{ON}$. 'B' dissolves in water to give a solution that turns litmus blue. Treatment of 'B' with acetic anhydride gives 'C', $\text{C}_8\text{H}_{17}\text{O}_2\text{N}$, which is insoluble in water and dilute bases, but soluble in dilute HCl.

'B' is found to be identical to the compound formed by the action of diethylamine on ethylene oxide.

28. When compound 'B' is treated with p – nitrobenzoyl chloride followed by moderate reduction with Ni/H_2 ; the compound formed is:



- (B)
- (C)
- (D)

29. Novocaine $\xrightarrow{\text{NaNO}_2, \text{HCl}}$ $\xrightarrow{\beta\text{-naphthol}}$ Product

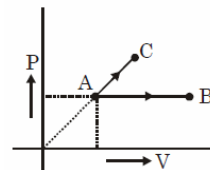
- (A)
- (B)
- (C)
- (D)

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. An organic compound **P** contains 62.07% carbon and 10.34% hydrogen and rest oxygen. Its vapour density is 29. This compound does not react with sodium metal, but its 2.9 g combines with X g of bromine (to give dibromo addition product). Find out value of (Y–X). [Where Y is total number of possible isomers (including tautomers also) of given organic compound P]
[Atomic mass: Br = 80]

31. One mole ideal monoatomic gas is heated in two different process according to path AB and AC. If temperature of state B and state C are equal. Calculate $10 \times \frac{q_{AC}}{q_{AB}}$



32. 6.84 gm $\text{Al}_2(\text{SO}_4)_3$ is needed to coagulate 2.5 L of As_2S_3 sol completely in 2.0 hrs. The coagulation value of $\text{Al}_2(\text{SO}_4)_3$ in terms of millimoles per litre is:
[Atomic mass: Al = 27, S = 32]

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

Treatment of 2, 4-pentanedione with KCN and acetic acid, followed by hydrolysis, gives two products, A and B. Both A and B are dicarboxylic acids of formula $\text{C}_7\text{H}_{12}\text{O}_6$. A melts at 98°C . When heated, B gives first a lactonic acid ($\text{C}_7\text{H}_{10}\text{O}_5$, m.p. 90°C) and finally a dilactone 'Q' ($\text{C}_7\text{H}_8\text{O}_4$, m.p. 105°C),
(A) What structure must B have that permits ready formation of both a monolactone and a dilactone?
(B) What is the structure of A?

33. 1mole of 'A' reacts with _____ moles of Na.
34. The degree of unsaturation in 'Q' is _____

Question Stem for Question Nos. 35 and 36

Question Stem

An ideal gas whose adiabatic exponents equal to 1.5 is expanded according to law

$P(\text{atm}) = 1.2 V(\text{Litre})$. The initial volume of the gas is equal to $\frac{1}{3}$ L. The final volume becomes 4 times the initial volume.

35. Increment in internal energy _____ litre atm
36. Magnitude of the work litre atm performed by the gas _____ litre atm.

Question Stem for Question Nos. 37 and 38

Question Stem

The reaction of Cl_2 gas with cold-dilute and hot-concentrated NaOH in water give sodium salts to two (different) oxoacids of chlorine, **P** and **Q**, respectively. The Cl_2 gas reacts with SO_2 gas, in presence of charcoal, to give a product **R**. **R** reacts with white phosphours to give a compound **S**. On hydrolysis, **S** gives an oxoacid of phosphours **T**.

37. The number of oxygen atoms in 'Q' is _____.
38. The number of atoms in a molecules 'R' is _____

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. The values of x for which $x^2 - 4x > \cot^{-1} x$ holds true belong to
 (A) $(-\infty, -1)$
 (B) $[5, \infty)$
 (C) $(-1, 5]$
 (D) None of these
40. From $(1, 2)$, tangents are drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$. Then
 (A) Sum of x -coordinates of points of contact is zero
 (B) Sum of x -coordinates of points of contact is 4
 (C) Sum of y -coordinates of points of contact is zero
 (D) Sum of y -coordinates of points of contact is 4
41. If $\int_a^b f(x)dx = \int_a^b f(x)dx$ ($a \neq b$) also $f(x) \neq 0$ for any $x \in (a, b)$ and $g(x) = \int_0^x f(x)dx$, then $\int_a^b f(x)g(x)dx$ when a and b are positive
 (A) Cannot be positive
 (B) Cannot be negative
 (C) Cannot be equal to zero
 (D) Nothing can be said
42. If $\alpha, \beta, \gamma \in \mathbb{R}$, then the determinant $\Delta = \begin{vmatrix} (e^{i\alpha} + e^{-i\alpha})^2 & (e^{i\alpha} - e^{-i\alpha})^2 & 4 \\ (e^{i\beta} + e^{-i\beta})^2 & (e^{i\beta} - e^{-i\beta})^2 & 4 \\ (e^{i\gamma} + e^{-i\gamma})^2 & (e^{i\gamma} - e^{-i\gamma})^2 & 4 \end{vmatrix}$ is
 (A) equal to 0
 (B) equal to $e^{i(\alpha+\beta+\gamma)}$
 (C) independent of α, β and γ
 (D) dependent of α, β and
43. If $AB = A$ and $BA = B$, where A and B are square matrices, then
 (A) $A^2 = A$
 (B) $A^2 = B$
 (C) $B^2 = B$
 (D) $B^2 = A$
44. If a complex number satisfies the equation $|z| - \log_2(z + 2) = 0$ ($\text{Re } z > 0$), then
 (A) $z = -\bar{z}$
 (B) $|z| = 2$
 (C) $\arg(z) = \frac{\pi}{2}$
 (D) $\arg(z) = 0$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

a, b, c, d is an A.P. with integer terms and $d = a^2 + b^2 + c^2$.

45. The common difference is
(A) 1
(B) 2
(C) 3
(D) none of these
46. The value of $a + 2b + 3c + 4d$ is
(A) 0
(B) 10
(C) 100
(D) none of these

Paragraph for Question Nos. 47 and 48

$f(x) = \sin 2\pi x + \{x\} : x \in [0, 10]$.

47. Number of points where f achieves local maximum is
(A) 10
(B) 11
(C) 20
(D) none of these
48. Number of roots of $f(x) = 0$ in $(0, 10)$ is
(A) 20
(B) 30
(C) 31
(D) none of these

Section – B (Maximum Marks: 12)

This section contains **THREE (03) questions**. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. f is a function such that $f(x) + 2f(1-x) = x^2 + 1$. Value of $f(3)$ is
50. Degree of differential equation of family of ellipses of same size having minor axis along a given line is
51. In $\triangle ABC$, $\tan A : \tan B : \tan C \equiv 1 : 2 : 3$. If $\sin A : \sin B : \sin C \equiv \sqrt{5} : 2\sqrt{2} : \lambda$, then $\lambda =$

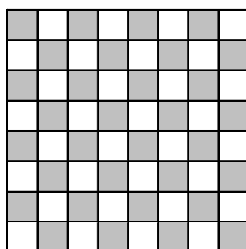
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Consider a chess board shown as



52. The number of ways in which 1 white and 1 black square can be selected such that they are not in same row or same column is _____
53. The number of ways in which a pair of black squares be chosen such that they have exactly one corner in common is _____

Question Stem for Question Nos. 54 and 55

Question Stem

$$\text{Let } \log_3 N = \alpha_1 + \beta_1$$

$$\log_5 N = \alpha_2 + \beta_2$$

$$\log_7 N = \alpha_3 + \beta_3$$

Where α_1, α_2 and α_3 are integers and $\beta_1, \beta_2, \beta_3 \in [0, 1)$.

54. Number of integral value of N if $\alpha_1 = 4$ and $\alpha_2 = 2$ _____
55. Difference of largest and smallest integral value of N if $\alpha_1 = 5, \alpha_2 = 3$ and $\alpha_3 = 2$.

Question Stem for Question Nos. 56 and 57

Question Stem

Let $f(x)$ and $g(x)$ be two differentiable functions, defined as:

$$f(x) = x^2 + xg'(1) + g''(2) \text{ and } g(x) = f(1)x^2 + xf'(x) + f''(x).$$

56. The value of $f(1) + g(-1)$ is _____
57. The number of integers in the domain of the function $F(x) = \sqrt{-\frac{f(x)}{g(x)}} + \sqrt{3-x}$ is _____